

AI or Overhiring? Explaining the Post-Pandemic Decline in Early-Career Hiring

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Abstract

Recent research finds that hiring among young workers has declined more in industries that are highly exposed to generative AI (Tucker, 2026). One explanation is that AI is reducing the need for some early-career workers. Another possibility is that many of these industries hired unusually fast during the COVID-19 pandemic and are now returning to more normal employment levels. This project will compare these two explanations using the U.S. Census Bureau's Quarterly Workforce Indicators, which provide data on hiring and employment by industry, state, age group, and quarter. I will estimate how much each state industry group's 2022 employment was above or below its pre-COVID trend, then examine whether AI exposure is still related to weaker hiring among workers ages 22 to 24. The goal is to examine whether the recent decline in early-career hiring is related to pandemic-era employment growth and AI exposure.

Purpose of the Study

This project aims to understand why early-career hiring has weakened in recent years. Tucker (2026) finds that workers ages 22 to 24 experienced a large decline in hiring in industry-state groups with greater exposure to AI after the introduction of ChatGPT. One possible explanation is that generative AI is changing the amount or the type of entry-level work that firms need.

However, the same period followed major changes in employment during the pandemic. Some industries expanded quickly from 2020 to 2022 and later experienced slower growth. Crane and Soto (2026), for example, find that employment in coder-intensive industries is currently consistent with a return toward its pre-COVID trend following unusually high pandemic employment.

My main research question is: After accounting for unusually high pandemic-era employment growth, is greater exposure to generative AI related to lower hiring among workers ages 22 to 24? The expected output will be a research paper supported by a cleaned dataset, Python code, a GitHub repository, statistical analysis, and visualizations.

Rationale

Understanding why early-career hiring is falling matters because early-career jobs are an important way for young workers to gain experience and begin their careers. If AI exposure remains related to lower hiring even after pandemic employment growth is taken into account, this may suggest that firms are changing how they use early-career workers. If the relationship between AI exposure and hiring becomes smaller after accounting for pandemic employment growth, then part of the decline may reflect a post-pandemic employment correction.

This project does not assume that either explanation is correct. Instead, it will consider both factors in the same analysis and examine how much each one is related to later hiring changes. This can provide a clearer view of recent labor market changes and may be useful for students, colleges, employers, and policymakers.

Literature Review

Researchers have developed several ways to measure how exposed different occupations are to generative AI. Eloundou et al. (2024) measure exposure based on the tasks workers perform and show that the potential effect of large language models differs widely across occupations. Tucker (2026) applies an AI exposure measure to industry-level Census data and finds that hiring for workers ages 22 to 24 fell more in highly exposed industries after ChatGPT's introduction.

At the same time, the evidence on recent labor market effects is not completely settled. Crane and Soto (2026) find that coder employment slowed sharply after the introduction of ChatGPT, but also show that employment in coder-intensive industries is close to the level expected from earlier industry trends. They describe this as being consistent with a return to trend after unusually high pandemic-era employment. Liu and Webber (2026) also find no broad decline in job postings among firms or industries with higher levels of AI adoption, showing that the labor market effects may differ across groups of workers rather than appearing equally across the economy.

My project will build on this research by making pandemic employment growth a part of the main analysis. Rather than claiming that previous studies have ignored this issue, I will test it directly by creating a measure of how far 2022 employment was above the level predicted from pre-COVID trends. I will then examine whether AI exposure is related to lower early-career hiring after this difference is taken into account. If time allows, I may also consider work-from-home exposure as an additional factor because the ability to work remotely differs greatly across occupations and industries (Dingel & Neiman, 2020).

Research Design

The main dataset for this project will be the U.S. Census Bureau's Quarterly Workforce Indicators (QWI). QWI provides information on employment, hiring, and worker separations by industry, state, age group, and quarter (U.S. Census Bureau, 2026). The planned unit of analysis is state by industry by age group by quarter. I plan to use four-digit NAICS industries if the data provide sufficient coverage at that level. My main outcome will be hiring among workers ages 22 to 24, since this group represents workers entering the labor market. I will also examine employment and separations to determine whether weaker employment outcomes reflect fewer workers being hired, more workers separating from their employers, or both.

I will combine the QWI data with an existing measure of occupational exposure to generative AI from Eloundou et al. (2024). Because AI exposure is measured by occupation while

QWI is organized by industry, I will use pre-pandemic occupation-industry employment shares to calculate an employment-weighted AI exposure score for each industry. This will allow me to compare industries with higher and lower exposure to generative AI.

I will also create a measure of pandemic employment growth above the pre-COVID trend. Using employment data from before COVID, I will estimate what employment in each industry would have been like by 2022 if its earlier trend had continued. I will then compare this predicted level with actual employment in 2022. The difference will provide a measure of how unusually high or low employment was before the widespread use of generative AI. I plan to use a simple pre-COVID trend model as the main approach because it is easy to interpret. If time allows, I will also use a time-series forecasting method as a robustness check to see whether the measure changes under a different way of estimating the expected employment path.

The analysis will first use descriptive analysis graphs and summary statistics to compare hiring trends across industries with different levels of AI exposure and pandemic employment growth. I will then use a difference-in-differences approach estimated through panel regression with fixed effects. This model will compare changes in early-career hiring before and after the introduction of ChatGPT across industries with different levels of AI exposure. I will interact AI exposure with a post-2022 indicator and then add a similar interaction for pandemic employment growth above trend. State-industry and quarter fixed effects will account for stable differences across groups and common changes over time. Standard errors will be clustered by industry. I will also use an event-study model to examine quarterly changes and check whether industries followed similar hiring trends before late 2022. If the data allow, I will extend the analysis to a triple-difference design comparing workers aged 22 to 24 with older workers in the same state-industry-quarter groups. All data cleaning, analysis, and visualization will be completed mainly in Python.

Preparation

I have experience using Python for data cleaning, visualization, regression analysis, and working with time-based data. These skills will help me collect, combine, and analyze the QWI and AI exposure datasets. To complete this project, I will need to strengthen my understanding of panel-data methods, including fixed effects, difference-in-differences, event studies, and clustered standard errors. I plan to learn these methods through course materials, econometrics textbooks, research papers, Python documentation, and guidance from Professor John Conzelmann in the Economics Department, whose research focuses on labor economics. I will also use GitHub to organize my code, document my data-processing decisions, and track changes throughout the semester.

Semester Work Plan

I will organize the project around the course deadlines. I will first complete the literature review and research design, then build the dataset and produce early results. Feedback from the early-results presentation, peer review, and meetings with professors will guide later revisions. The

final weeks will focus on completing the paper, documenting the code, writing the executive summary, and preparing the poster presentation.

Dates	Tasks
Sept. 14 to Sept. 20	Complete the literature review, strengthen the research question and literature gap
Sept. 21 to Oct. 4	Finalize the difference-in-differences design, define the main variables and fixed effects, and meet with professors for comments
Oct. 5 to Oct. 18	Download and clean the QWI and AI exposure data, create the pandemic employment growth measure, estimate the first models, and submit early results
Oct. 19 to Nov. 1	Present the early results, use feedback to improve the data and models, and begin writing the methods and results sections
Nov. 2 to Nov. 15	Complete the fixed-effects and event-study analysis, submit the first paper draft
Nov. 16 to Dec. 6	Meet with professors, complete feasible robustness checks, and revise the paper
Dec. 7 to Dec. 17	Incorporate final feedback, finish the paper and document the Python code, submit the final draft, and prepare the poster presentation

References

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